Quantitative Cutoffs and Physical Gram Certificates for the BFSS Threshold-State Argument

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Abstract

The assigned manuscript claims a unique normalizable zero-energy state of the relative SU(N) BFSS model for every finite $N \geq 2$. We analyze its closed supercharges, massive index, cluster profiles, and ordered compactness arguments without promoting finite residuals to a proof of that claim. A quantitative version of its rotation anticommutator gives an explicit odd-parity residual–tail obstruction without a position moment assumption. For its whole-block recovery map, an exact truncated second-moment identity yields conditional charge-squared rates $h^{-s/3}$, $h^{-2/3}\log h$, and $h^{-2/3}$ under a squared-tail envelope of order $s < 2$, $s = 2$, and $s > 2$, respectively. Matrix-valued tails retain correlations in a finite family, and the actual massive spectral gap converts full charge errors into a physical Gram certificate. These are quantitative cutoff and conditioning refinements, not an independent index calculation, tightness theorem, or construction of a BFSS state. Contribution assessment: C2/5, modest source-specific residual–tail and recovery bounds with explicit physical Gram budgets, conditional on the source operator identities and massive gap rather than a new threshold-state theorem.

Contents

1 The Operator and the Scope of the Claim **2**

2 Source Architecture and Original Comparisons **3**

3 An Explicit Odd-Parity Residual–Tail Bound **4**

4 Whole-Block Recovery with a Tail Modulus **6**

5 A Matrix Tail and a Finite Massive-Kernel Certificate **8**

6 Benefits, Costs, and Unfinished Global Obligations **9**

7 Evidence and Contribution Boundary **10**

1 The Operator and the Scope of the Claim

Fix a finite integer $N \geq 2$, put $d_N = N^2 - 1$ and $d = 9d_N$, and give traceless Hermitian matrices the trace inner product. For an orthonormal color basis T_A, use the real bracket $[X, Y]_{\mathrm{r}} = (XY - YX)/i$ and coordinates x_i^A, $1 \leq i \leq 9$. Write $p_i^A = -i\partial_{x_i^A}$ and $r = |x| = (\sum_{i,A}(x_i^A)^2)^{1/2}$. The fermion operators satisfy

$$\{\theta_\alpha^A, \theta_\beta^B\} = \delta_{\alpha\beta}\delta_{AB}\mathrm{I}, \qquad \alpha, \beta = 1, \ldots, 16.$$

The irreducible complex Clifford module is denoted $\mathcal{F}_N$. Real symmetric 16×16 matrices γ^i satisfy $\{\gamma^i, \gamma^j\} = 2\delta_{ij}\mathrm{I}$; set $\gamma^{ij} = [\gamma^i, \gamma^j]/2$ and $T = \gamma^1\gamma^2\gamma^3$. All Hilbert inner products are conjugate linear in their first argument.

The physical space and its undeformed realization are

$$\mathscr{H} = \left[L^2(\mathbb{R}^d; \mathcal{F}_N)\right]^{\mathrm{SU}(N)}, \qquad H_N = \text{the nonnegative operator of the closed supercharge form.} \tag{1}$$

Gauge invariance acts simultaneously on the coordinates and Clifford fiber. It is not invariance of a scalar coordinate function alone. The center-of-mass U(1) multiplet is absent. There is no imposed boundary at a finite radius or at a valley wall.

For a real unit spinor v, the source's two-parameter charge is

$$D_v(h, m) = \sum_i \theta(\gamma^i v) \cdot p_i - h \sum_{i<j} \theta(\gamma^{ij} v) \cdot [x_i, x_j]_{\mathrm{r}} + mB_v(x), \tag{2}$$

$$B_v(x) = \sum_i s_i \theta(\gamma^i T v) \cdot x_i, \qquad (s_1, s_2, s_3, s_4, \ldots, s_9) = (2, 2, 2, -1, \ldots, -1). \tag{3}$$

Here $h > 0$, $m \geq 0$, and $\theta(a) \cdot x = \sum_{A,\alpha} \theta_\alpha^A a_\alpha x^A$. For an orthonormal real spin basis $(v_\nu)_{\nu=1}^{16}$ let

$$\mathfrak{q}_{h,m}(\psi) = \frac{1}{16} \sum_{\nu=1}^{16} \left\| D_{v_\nu}(h, m)\psi \right\|^2 . \tag{4}$$

The form domain is the simultaneous graph completion of the gauge-invariant smooth compactly supported core. At $m = 0$ the gauge square is independent of the unit spinor; H_N corresponds to $(h, m) = (1, 0)$. The source proves essential self-adjointness using the constant elliptic symbol, local H^1 regularity, radial graph cutoffs, and smoothing. We use that realization, not a formal differential expression with an unspecified domain. In particular,

$$[D_v, \chi] = -i \sum_i \theta(\gamma^i v) \cdot \partial_i \chi, \qquad \|[D_v, \chi](x)\|_{\mathrm{op}} = \frac{|\nabla\chi(x)|}{\sqrt{2}}. \tag{5}$$

The exterior grading uses the T-polarized module with empty vector minimizing $i\theta^t T\theta/2$. Write Π_- for odd parity. Choose the source's real unit spinor u satisfying $\gamma^{12}u = \gamma^{45}u = \gamma^{67}u = \gamma^{89}u$. Let G be the connected torus with Lie algebra $\{t_1E_{12} + t_2E_{45} + t_3E_{67} + t_4E_{89} : \sum_j t_j = 0\}$, which fixes u, and put $\mathscr{H}_G = \mathscr{H}^G$. The source establishes the form comparison

$$\mathfrak{q}_{h,m}(\psi) \leq \tfrac{3}{2} \left\| D_u(h, m)\psi \right\|^2 \quad (\psi \in \mathscr{H}_G \cap D(D_u(h, m))). \tag{6}$$

It is a comparison on this specified invariant space. For a general gauge vector the averaged form, rather than one chosen charge, is the safe residual norm.

1.1 What is imported and what is proved here

The assigned preprint [1] asserts $\dim \mathcal{K}_N = 1$, where $\mathcal{K}_N = \operatorname{Ker} H_N$, and that its vectors are Spin(9) singlets and even. All scientific modules were freshly read, but reading is not an independent verification of every global compactness or index step. This report proves the quantitative statements in Sections 3–5 from explicit operator inputs. It does not independently establish the existence, finite dimensionality, or uniqueness of $\mathcal{K}_N$. In particular, the positive *massive* gap used below is not a positive gap at the undeformed threshold.

2 Source Architecture and Original Comparisons

The following dependency order is substantive. Interchanging it can silently assume the state whose existence is at issue.

1. **Algebra and domains.** Section 2 of the source fixes gauge terms in the square, the mass deformation, the torus comparison, and a rotation anticommutator. Its radial cutoff argument proves parity of any existing massless kernel vector without assuming a position moment or finite kernel dimension.

2. **Massive index at small coupling.** Section 3 proves confinement and compact graph embedding for each $h > 0, m = 1$. Its normal wells are the $p(N)$ partition-labeled $\mathfrak{su}(2)$ orbits. Oscillator energies, stabilizer characters, and parity are all retained. Norm exhaustion in normal slices and recovery of every vacuum produce the Fredholm index $p(N)$; local normal oscillators alone do not.

3. **Separated centers.** Section 4 uses the actual unitary slice density $V_C^{1/2} |\det K_e(b)|^{1/2} h^{-d_f/4}$, not an unweighted quotient. Central-torus neutrality cancels angular terms. The fast kernel is an even line with a controlled reduced inverse near compact separated-center sets, including nearby noncommuting blocks.

4. **Graph test lifts.** Section 5 corrects the oscillator product by $-h^{-1/2} E_v^{-1}(1 - P_v) A f_h$. Projection alone leaves an order-one transverse output. Reduced-inverse regularity, internal kinetic/bracket estimates, chart compatibility, and positive orbit normalization justify the corrected output convergence.

5. **Exterior bounds and profiles.** Section 6 inducts jointly in block size. Smaller-block exterior estimates exclude intermediate scales, giving exact proper-stratum norm densities. Corrected tests pass charges by duality. Collision sets have codimension nine and cutoff cost $O(\varepsilon^7)$. Center Hardy inequalities then prove the current exterior estimate before assuming anything about $\mathcal{K}_N$.

6. **Large-coupling count.** Section 7 excludes massive escape, exhausts polarized norms, and obtains an odd massive gap. Supersymmetric pairing gives a gap off the massive kernel. There are $p(N) - 1$ proper partition lines; arbitrary finite lists in the possible whole-collapse kernel can be recovered. Projection and Gram rank finish the claimed count.

Two details merit preservation. Compactness is applied jointly in all center/internal variables with finite Clifford fiber, not to an uncontrolled infinite-dimensional spectator. Also, only equal-sized block permutations are divided out in a labeled chart, with graded fiber transport and the normalized fast vacuum; no extra multiplicity or sign may be inserted into the count.

2.1 Fresh original Ram inputs

The original Volume 21 chapters on state domains, physical metrics, and spectral limits [5, 6, 7] were read directly. Their graph-core arguments distinguish a closed realization from a formal formula. Their synthesis identity $G = J^*J$ and orthogonal residual decomposition retain both the physical metric and omitted output. Their explicit spectral-pollution example shows why nested graph-core trial spaces alone do not establish a spectral conclusion. The report uses these actual deductions, not prior comparative reports, as the Ram framework for the finite-family certificate below. The Hilbert-space identities and layer-cake estimates remain classical; they are not claimed as historically new transforms.

2.2 Public status checked on October 9, 2026

Lin's primary arXiv record remains version 3, August 18, 2026. Its Section 3.1 distinguishes the $N = 2$ index-and-parity conclusion from the general finite-N bound-state expectation [3]. This report does not infer consensus from the assigned September preprint's title. Hasler–Hoppe's Theorem 1(a), anticommutator Lemma 2(c), and cutoff proof already establish the rotational-invariance mechanism for possible zero-energy states [2]; our finite-residual bound is a quantitative use of that mechanism. The primary abstract of Boulton–Garcia del Moral–Restuccia explicitly concerns outer Dirichlet problems on valleys [4]. That problem is not identical to the whole-space kernel in (1). The external access scope is recorded separately: these selected passages and metadata were read, not all cited external proofs. No priority or exhaustive literature claim is made.

3 An Explicit Odd-Parity Residual–Tail Bound

Let $F = E_{12}$ be the real skew matrix with $F_{12} = 1, F_{21} = -1$. The source rotation generator and its odd primitive are

$$J(F) = p \cdot Fx + \tfrac{i}{2}\theta^t S(F)\theta, \qquad S(F) = \tfrac{1}{4}\sum_{ij} F_{ij}\gamma^i\gamma^j, \tag{7}$$

$$M_v(F) = \sum_i \theta(P_i(F)v)\cdot x_i, \qquad P_i(F) = \tfrac{1}{16}\sum_j F_{ji}\gamma^j + \tfrac{1}{112}\sum_{jk} F_{jk}\gamma^{ijk}. \tag{8}$$

Here γ^{ijk} is the completely antisymmetric Clifford product. The identity on the physical smooth core is

$$\sum_{\nu=1}^{16}\{D_{v_\nu}(h,0), M_{v_\nu}(F)\} = J(F). \tag{9}$$

This is the source's algebraic input. Its normalization can be checked by $\mathrm{Tr}(\gamma^j P_i) = F_{ji}$, $\mathrm{Tr}(\gamma^{jk}P_i) = 0$, and $\sum_i \gamma^i P_i^t = -S(F)/2$.

Lemma 3.1 (Pointwise primitive budgets)**.** *For $M_\nu = M_{v_\nu}(E_{12})$ and any fiber vector w,*

$$\sum_\nu \|M_\nu(x)w\|^2 = \left(\frac{|x_1|^2+|x_2|^2}{32} + \frac{\sum_{i=3}^9 |x_i|^2}{392}\right)\|w\|^2 \le \frac{r^2}{32}\|w\|^2, \tag{10}$$

$$\sum_\nu \|B_{v_\nu}(x)w\|^2 = 8\sum_i s_i^2|x_i|^2\|w\|^2 \le 32r^2\|w\|^2. \tag{11}$$

For a single unit v, $\|B_v(x)\|_{\mathrm{op}} \le \sqrt{2}r$.

Proof. For a real Clifford coefficient vector a, the CAR give $\theta(a)^2 = |a|^2\mathrm{I}/2$. In (8), $P_1 = -\gamma^2/16$, $P_2 = \gamma^1/16$, and $P_i = \gamma^{i12}/56$ for $i \geq 3$. Their trace Gram matrix is diagonal with entries $1/16, 1/16, 1/196, \ldots, 1/196$. Sum $|\sum_i x_i^A P_i v_\nu|^2/2$ over ν, A to obtain (10). For the mass primitive, the symmetric part of $\langle \gamma^i T v, \gamma^j T v\rangle$ is δ_{ij} for a real unit v. Thus $B_v(x)^2 = (\sum_i s_i^2 |x_i|^2)\mathrm{I}/2$. Summing sixteen copies proves (11) and the single-charge bound. □

Theorem 3.2 (Odd-sector cutoff obstruction)**.** *For every $h > 0, m \geq 0$, let ψ be a normalized odd gauge-invariant vector in the form domain of (4). Put $\epsilon = \mathfrak{q}_{h,m}(\psi)^{1/2}$ and $T_R = \left\|\mathbf{1}_{\{r>R\}}\psi\right\|^2$. For every $R > 0$,*

$$1 \leq 2\sqrt{2}\pi R\epsilon + 8\pi m R^2 + (1+\pi)T_R. \tag{12}$$

No assumption $r\psi \in L^2$ is required. On $\mathscr{H}_G$, the first term may be replaced by $2\sqrt{3}\pi R\,\|D_u(h,m)\psi\|$.

Proof. First use a smooth compact physical vector. Let χ_R equal one on $r \leq R$, decrease linearly to zero on $[R, 2R]$, and vanish thereafter. Lipschitz multiplication is valid by local H^1 regularity; smooth approximations give the same estimates. Thus $|\nabla\chi_R| \leq 1/R$ and $r|\nabla\chi_R| \leq 2$. Let $U(t)$ be the 12-plane spin rotation with generator $J(F)$, with either consistent sign convention. The central element reached at $t = 2\pi$ acts as exterior parity, so $U(2\pi)\psi = -\psi$. Both the averaged massive form and the radial tail are rotation invariant. For $f(t) = \langle \psi, \chi_R U(t)\psi\rangle$ the endpoints differ by $2\langle\psi, \chi_R\psi\rangle$ in absolute value.

Write $D_\nu^0 = D_{v_\nu}(h, 0)$ and $D_\nu^m = D_\nu^0 + mB_{v_\nu}$. Using (9), moving one charge across the scalar cutoff, and putting $\zeta = U(t)\psi$, gives two residual terms, two mass terms, and one cutoff commutator term. The two residual terms are bounded by

$$\left(\sum_\nu \|D_\nu^m\psi\|^2\right)^{1/2}\left(\sum_\nu \|\chi_R M_\nu \zeta\|^2\right)^{1/2} + \left(\sum_\nu \|\chi_R M_\nu\psi\|^2\right)^{1/2}\left(\sum_\nu \|D_\nu^m\zeta\|^2\right)^{1/2} \leq 2\sqrt{2}R\epsilon.$$

Pointwise (10)–(11) bound each mass term by $m\int \chi_R r^2|\psi||\zeta| \leq 4mR^2$. For the commutator, the pointwise spin sum from (5) is $8|\nabla\chi_R|^2|\psi|^2$. Hence its bound is

$$\frac{1}{2}\int r|\nabla\chi_R|\,|\psi|\,|\zeta| \leq \|\mathbf{1}_{r>R}\psi\|\,\|\mathbf{1}_{r>R}\zeta\| = T_R.$$

Consequently $|f'(t)| \leq 2\sqrt{2}R\epsilon + 8mR^2 + T_R$. Integrate over length 2π and divide by two. Since $1 \leq \langle\psi, \chi_R\psi\rangle + T_R$, this proves (12). Finally use the gauge-invariant core, parity projection, and normalization to approximate an arbitrary form-domain vector. Form norms and cutoff tails converge; all constants are independent of the approximation. Equation (6) proves the last assertion. □

For $m = 0$ and a certified $T_R \leq \tau < (1+\pi)^{-1}$, the theorem gives

$$\mathfrak{q}_{h,0}(\psi)^{1/2} \geq \frac{1 - (1+\pi)\tau}{2\sqrt{2}\pi R}. \tag{13}$$

Thus uniformly tight normalized odd sequences cannot have averaged charge residual tending to zero. This is not a uniform gap on all odd massless states: R and the tail hypothesis are part of the bound. For $m > 0$, usefulness additionally requires $8\pi mR^2$ to be small. In particular (12) is not a replacement for the source's large-h, fixed-$m = 1$ odd-gap proof, which treats separate cluster profiles.

Corollary 3.3 (A mixed-parity estimate)**.** *For any form-domain vector ψ, set $n = \|\Pi_-\psi\|$, $e = \mathfrak{q}_{h,m}(\Pi_-\psi)^{1/2}$, and $t = \|\mathbf{1}_{r>R}\Pi_-\psi\|^2$. Then*

$$(1 - 8\pi m R^2)n^2 \le 2\sqrt{2}\pi R e n + (1+\pi)t.$$

If $a = 1 - 8\pi m R^2 > 0$, this implies

$$n \le \frac{b + \sqrt{b^2 + 4a(1+\pi)t}}{2a}, \qquad b = 2\sqrt{2}\pi R e.$$

Moreover $e \le \mathfrak{q}_{h,m}(\psi)^{1/2}$ and $t \le \|\mathbf{1}_{r>R}\psi\|^2$.

Proof. Apply Theorem 3.2 to $\Pi_-\psi/n$ if $n > 0$ and multiply by n^2; the case $n = 0$ is immediate. Charges reverse parity, so the even and odd input contributions to their squared norms are orthogonal. Radial cutoffs commute with parity. Solve the resulting quadratic. □

4 Whole-Block Recovery with a Tail Modulus

The source's whole-recovery lemma starts with a vector already in $\mathcal{K}_N$, not with an arbitrary numerical near-null vector. Its purpose is to transfer an arbitrary finite orthonormal list into the massive space for a dimension argument. The next estimates have exactly that direction of implication.

For a normalized $\phi \in \mathcal{K}_N$, define the squared radial tail and truncated second moment by

$$F(R) = \int_{|y|>R} |\phi(y)|^2 \,\mathrm{d}y, \qquad I_2(R) = \int_{|y|\le R} |y|^2 |\phi(y)|^2 \,\mathrm{d}y. \tag{14}$$

No probability interpretation beyond the normalized physical norm is needed. Fix $B > 0$ and take the radial cutoff χ_B used in the previous section. Let

$$(S_h\phi)(x) = h^{d/6}\phi(h^{1/3}x), \qquad z_{h,B} = \chi_B S_h \phi, \qquad L = Bh^{1/3}. \tag{15}$$

The Jacobian $h^{-d/3}$ makes S_h unitary, and direct scaling gives $D_v(h,0)S_h = h^{1/3}S_h D_v(1,0)$ distributionally.

Proposition 4.1 (Exact tail account and charge bound)**.** *For every $h, B > 0$ and every real unit spinor v, the vector $z_{h,B}$ belongs to the charge domain and*

$$1 - F(L) \le \|z_{h,B}\|^2 \le 1, \tag{16}$$

$$\|D_v(h,1)z_{h,B}\| \le \frac{\sqrt{F(L)}}{\sqrt{2}B} + \sqrt{2}h^{-1/3}\sqrt{I_2(2L)}, \tag{17}$$

$$\|D_v(h,1)z_{h,B}\|^2 \le B^{-2}F(L) + 4h^{-2/3}I_2(2L). \tag{18}$$

The exact scalar identity is

$$I_2(R) = 2\int_0^R rF(r)\,\mathrm{d}r - R^2F(R). \tag{19}$$

For fixed B, (17) tends to zero for every such $\phi \in L^2$, even if $I_2(\infty) = \infty$.

Proof. The norm loss is supported where $|y| > L$. The massless output is exactly $[D_v, \chi_B]S_h\phi$, whose norm is bounded by the first term of (17). The mass contribution is bounded by $\sqrt{2}\,\|rz_{h,B}\| \le \sqrt{2}h^{-1/3}\sqrt{I_2(2L)}$ by Lemma 3.1. Both outputs are in L^2; the source's maximal-equals-minimal domain result places the vector in the actual graph domain. The triangle inequality and $(a+b)^2 \le 2a^2 + 2b^2$ give the displayed estimates.

Tonelli applied to $\min(|y|^2, R^2) = \int_0^R 2r\mathbf{1}_{|y|>r}\,\mathrm{d}r$ gives $I_2(R) + R^2F(R) = 2\int_0^R rF(r)\,\mathrm{d}r$. Radial spheres have zero Lebesgue measure, so the endpoint convention is immaterial here. For fixed B, $h^{-2/3}|y|^2\mathbf{1}_{|y|\le 2Bh^{1/3}} \le 4B^2$ and tends pointwise to zero. Dominated convergence against $|\phi|^2$ proves $h^{-2/3}I_2(2L) \to 0$. Also $F(L) \to 0$. □

Corollary 4.2 (Conditional recovery rates)**.** *Assume in addition $F(r) \le Ar^{-s}$ for all $r \ge 1$, with given $A > 0, s > 0$. For $R \ge 1$ set*

$$\Theta_s(R) = \begin{cases} 1 + \dfrac{2A(R^{2-s} - 1)}{2 - s}, & s \ne 2, \\ 1 + 2A\log R, & s = 2. \end{cases} \tag{20}$$

Whenever $L = Bh^{1/3} \ge 1$,

$$\|D_v(h,1)z_{h,B}\|^2 \le AB^{-2}L^{-s} + 4h^{-2/3}\Theta_s(2L). \tag{21}$$

With A, s, B fixed this is $O(h^{-s/3})$ for $0 < s < 2$, $O(h^{-2/3}\log h)$ for $s = 2$, and $O(h^{-2/3})$ for $s > 2$. These are bounds for the squared *charge norm; the charge itself has their square-root orders.*

Proof. Use $F \le 1$ on $(0,1)$, the assumed envelope on $[1,R]$, and discard the nonnegative subtraction in (19). This gives $I_2(R) \le \Theta_s(R)$. Substitute into (18). For $s > 2$, $\Theta_s(R) \le 1 + 2A/(s-2)$; the other two cases follow by integration. No uniformity as $s \to 2$ is asserted by the separate big-O statements; the explicit formula handles that transition. □

4.1 Comparison with a single split and the absence of a free rate

The source proves convergence by splitting at a physical $\eta > 0$, bounding the mass second moment by $\eta^2 + (2B)^2F(h^{1/3}\eta)$, then taking $h \to \infty$ before $\eta \downarrow 0$. Under the additional power envelope, optimizing only that split yields order $h^{-2s/[3(s+2)]}$ for the squared mass norm. Integrating all radii in (19) instead gives the three orders in Corollary 4.2. This is an improvement of the *conditional quantitative estimate*, not a correction of the source's valid moment-free convergence proof. Neither its L^2 hypothesis nor a formal cluster asymptotic supplies the envelope Ar^{-s}.

For a precise model of the threshold at $s = 2$, take a radial norm law with density $sr^{-s-1}\,\mathrm{d}r$ on $r \ge 1$. It can be realized by an $L^2(\mathbb{R}^d)$ scalar function by dividing this density by the spherical Jacobian. Then $F(R) = R^{-s}$ for $R \ge 1$ and

$$I_2(R) = \begin{cases} \dfrac{s(R^{2-s} - 1)}{2 - s}, & s \ne 2, \\ 2\log R, & s = 2. \end{cases} \tag{22}$$

For a cutoff equal to one up to L, the retained position moment is at least $h^{-2/3}I_2(L)$, so the three regimes cannot be improved using only this tail class. These are scalar norm-law fixtures, not BFSS kernel vectors or sharpness claims for BFSS states.

More basically, for any fixed h an L^2 unit vector may be supported outside $|y| > 2Bh^{1/3}$, and the cutoff then retains nothing. Thus uniform recovery over the full L^2 unit ball is false. A fixed vector may put arbitrarily slowly decaying amounts of norm on increasingly distant annuli. No numerical algebraic rate follows from its norm alone.

5 A Matrix Tail and a Finite Massive-Kernel Certificate

Let $\phi_1, \dots, \phi_q$ be any finite orthonormal list in $\mathcal{K}_N$; its existence is a premise, not a conclusion. Set $\Phi c = \sum_{a=1}^{q} c_a \phi_a$ and define Hermitian matrices

$$T(R)_{ab} = \int_{|y|>R} \langle \phi_a(y), \phi_b(y) \rangle \, \mathrm{d}y, \qquad U_2(R)_{ab} = \int_{|y|\le R} |y|^2 \langle \phi_a(y), \phi_b(y) \rangle \, \mathrm{d}y. \tag{23}$$

Then $0 \preceq T(R) \preceq \mathrm{I}$ and the exact matrix layer-cake identity holds entrywise, or by polarization of (19):

$$U_2(R) = 2 \int_0^R r T(r) \, \mathrm{d}r - R^2 T(R). \tag{24}$$

Put $W = \chi_B S_h \Phi$, $G_W = W^* W$, and $E_W = (D_u(h,1)W)^*(D_u(h,1)W)$, with whole-space inner products.

Proposition 5.1 (Correlated recovery budgets)**.** *For $L = Bh^{1/3}$,*

$$G_W \succeq \mathrm{I} - T(L), \qquad E_W \preceq B^{-2} T(L) + 4h^{-2/3} U_2(2L). \tag{25}$$

If $T(r) \preceq A r^{-s} \mathrm{I}$ for $r \ge 1$, the right side of the second inequality is at most the scalar bound (21) times I*. Individual estimates $F_a(r) \le A_0 r^{-s}$ imply this premise with $A = qA_0$ by the trace bound; replacing it by A_0 without a uniform span estimate is invalid.*

Proof. Evaluate every assertion at $c \in \mathbb{C}^q$. The norm loss is $\int (1 - \chi_B(h^{-1/3}y)^2) |\Phi c(y)|^2 \, \mathrm{d}y \le c^* T(L) c$. Apply the proof of (18) to Φc without normalizing it to get the second matrix inequality. For a positive matrix its largest eigenvalue is at most its trace, proving the last assertion. □

The next proposition applies also to the source's combined list of proper-cluster lifts and whole-block recoveries. Their Gram matrix must include their actual cross terms. Asymptotic orthogonality without a finite error bound cannot be replaced by an exactly diagonal Gram matrix.

Proposition 5.2 (Physical Gram certificate at a massive gap)**.** *Let D be a self-adjoint operator on a Hilbert space and P its kernel projection. Assume, for a supplied $\delta > 0$,*

$$\|(1-P)z\| \le \delta^{-1} \|Dz\| \quad (z \in D(D)). \tag{26}$$

Let $W : \mathbb{C}^q \to D(D)$ be a finite synthesis map and put $G = W^*W$, $E = (DW)^*(DW)$. Then

$$(PW)^*(PW) = G - ((1-P)W)^*((1-P)W) \succeq G - \delta^{-2} E. \tag{27}$$

Consequently $G - \delta^{-2}E \succ 0$ implies $\dim \operatorname{Ker} D \ge q$. If $G \succ 0$, a coordinate-invariant sufficient condition is

$$\rho = \lambda_{\max}(G^{-1/2} E G^{-1/2}) < \delta^2, \qquad (PW)^*(PW) \succeq (1 - \rho/\delta^2) G. \tag{28}$$

Certified matrices $G \succeq \widehat{G} - \Delta_G$ and $E \preceq \widehat{E} + \Delta_E$ may replace the exact matrices if $\Delta_G, \Delta_E \succeq 0$ are whole-space error bounds.

Proof. The two projected parts of Wc are orthogonal, proving the equality. Square (26) for Wc and vary c to obtain the form inequality. A positive projected Gram matrix is injective synthesis into the kernel. Conjugating by $G^{-1/2}$ proves (28); the asserted error replacement follows by the same Loewner order. An invertible coefficient change $W \mapsto WC$ congruently changes all three physical Gram matrices, so positivity and the generalized residual ratio do not depend on the coordinate choice. □

In the application, $D = D_u(h,1)|_{\mathscr{H}_G}$. The source's large-coupling kernel-gap proposition supplies (26) for all sufficiently large h, with some N-dependent $\delta > 0$. This report neither computes that δ nor its threshold in h. The source proves every element of $\mathcal{K}_N$ is a singlet and even, so radial whole-recovery vectors lie in $\mathscr{H}_G$; using that assertion is explicitly part of the interface. Combining the propositions gives the concrete sufficient condition

$$\mathrm{I} - T(L) - \delta^{-2}\big(B^{-2}T(L) + 4h^{-2/3}U_2(2L)\big) \succ 0. \tag{29}$$

It is a conditional finite-h refinement of the source's eventual Gram positivity, not a way to infer its massive gap from finite eigenvalues.

5.1 Worked budget and strict-boundary examples

Suppose a proposed finite orthonormal kernel family has the uniform matrix envelope $T(r) \preceq r^{-1}\mathrm{I}$ for $r \geq 1$, and suppose the actual massive gap has been certified with $\delta = 1$ at $h = 10^6$. Choose $B = 1$, so $L = 100$. Then $T(L) \preceq \mathrm{I}/100$ and $\Theta_1(200) = 399$. Formula (29) yields

$$(PW)^*(PW) \succeq \left(1 - \frac{1}{100} - \frac{1}{100} - \frac{4 \cdot 399}{10^4}\right)\mathrm{I} = \frac{2051}{2500}\mathrm{I}.$$

Every number is an exact rational calculation. No BFSS tail envelope or gap of this size is claimed; the example shows what supplied bounds would certify, and what data are still missing.

Strictness in (28) is essential. On $\mathbb{C}^2$ let $D = \mathrm{diag}(0,\delta)$ and $w = ae_0 + be_1$. Then $G = |a|^2 + |b|^2$, $E = \delta^2|b|^2$, and $G - \delta^{-2}E = |a|^2 = \|Pw\|^2$. At $a = 0$ the boundary equality holds and the projected vector is zero. This fixture also shows that the subtraction in (27) cannot be uniformly reduced under only the stated gap hypothesis.

For a full-residual warning take $D = \mathrm{diag}(-1,1)$ and $w = (1,1)/\sqrt{2}$. Its compressed expectation is zero but $\|Dw\| = 1$ and $\mathrm{Ker}\, D = 0$. For a distinct threshold warning take $D = -i\partial_x$ on $H^1(\mathbb{R})$ and a unit smooth compactly supported f. Then $f_L(x) = L^{-1/2}f(x/L)$ has $\|Df_L\| = L^{-1}\,\|f'\| \to 0$, but D has no L^2 kernel and these vectors are not tight. The examples are classical operator diagnostics, not BFSS counterexamples.

6 Benefits, Costs, and Unfinished Global Obligations

The cutoff rotation bound makes a useful negative test: a declared odd state cannot simultaneously have a small full averaged charge and a small physical tail at an incompatible radius. Its constants are independent of color dimension in the stated trace/CAR normalization. That uniform local constant does not make the source's global thresholds or cluster gaps uniform in N.

The recovery estimate uses an integral of the actual tail, rather than paying for only one arbitrary split. It preserves the source's moment-free limit and gives rates only when an extra envelope is certified. Matrix tails may reduce the cost of treating each column separately, but estimating them requires correlations over the entire configuration space. A finite set of moments can describe a finite trial family; it cannot establish a uniform tail bound on the unknown continuum kernel. No computational complexity or speedup is measured.

The following global obligations remain those of the assigned source: the massive index including stabilizer characters; regularity of fast projections and reduced inverses with the correct gauge density; graph convergence of the corrected lifts; all intermediate-scale and massive escape exclusions; polarized norm exhaustion; the parity and supersymmetric pairing steps; and the recovery/partition count connecting these inputs. Small residuals, a finite Gram determinant, or an index number

alone do not replace this chain. In particular an index fixes a signed difference, and it counts a kernel dimension only after the opposite parity and positive-spectrum pairing have been controlled.

Three further developments are plausible but conditional. First, a proved global squared-tail bound for a specified BFSS kernel family would activate (21); a leading asymptotic on one separated stratum does not suffice for all nested collisions. Second, effective constants in the source's massive odd-gap argument, together with certified proper-profile lift errors, would turn its final eventual Gram argument into an explicit finite-h test. Third, one could optimize the cutoff transition rather than fixing $[B, 2B]$, provided its gradient cost and the full retained mass moment are both kept. These are research tasks, not consequences already supplied here.

7 Evidence and Contribution Boundary

The literal-read record covers all 11 cached public text files, including every scientific section, the main file, induction diagram source, bibliography, and README. The three freshly read original private chapters have 164, 210, and 242 lines. Their included figures and cited reports were not thereby read. Pre-read hashes were persisted before displays and matched after reading. Captured-tree authentication is provenance, not a live-branch or proof-validity statement.

The owned executable checks use rational arithmetic for Clifford-word trace budgets, layer-cake identities, conditional rates, and physical projection/Gram fixtures: 81 exact parameter cases and 742 assertions, with zero numerical or foreign/parent cases. Code hashes are recorded in the owned test receipt; these are checks of local algebra, not samples of unknown BFSS ground states or executions of an upstream proof checker. The local arguments are written out above, and the global source conclusions remain imported even though its scientific text was read. No mathematical correction to the source is asserted. Applicability remains L3. C2 reflects a useful but modest quantitative elaboration of existing cutoff, tail, and Hilbert-space methods, without a new global bound-state result or a historical novelty claim.

References

[1] OpenAI, *The unique threshold bound state of the* SU(N) *BFSS model*, September 24, 2026, assigned cached preprint, captured commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Publisher source directory. All seven scientific sections freshly read; their global conclusion is not independently certified by this report.

[2] D. Hasler and J. Hoppe, *Zero Energy States of Reduced Super Yang–Mills Theories in* $d+1 = 4, 6$ *and 10 dimensions are necessarily* Spin(d) *invariant*, arXiv:hep-th/0211226v1, 2002. Theorem 1(a), Lemma 2(c), and the cutoff argument on pp. 3–5 inspected.

[3] H. W. Lin, *TASI lectures on Matrix Theory from a modern viewpoint*, arXiv:2508.20970v3, August 18, 2026, Section 3.1 and version metadata inspected October 9.

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[5] M. Subbarao, *Ram Calculus*, Volume 21, original chapter `ch01a-state-domain.tex`, lines 1–164, confidential local corpus. Closed graphs, transported domains and completion.

[6] M. Subbarao, *Ram Calculus*, Volume 21, original chapter `ch01d-physical-metrics.tex`, lines 1–210, confidential local corpus. Physical Gram matrices, omitted residuals, relative metric accuracy and finite measurement limitations.

[7] M. Subbarao, *Ram Calculus*, Volume 21, original chapter `ch01f-spectral-limits.tex`, lines 1–242, confidential local corpus. Explicit spectral pollution and graph-domain inversion bounds. These originals were read locally and were not uploaded.